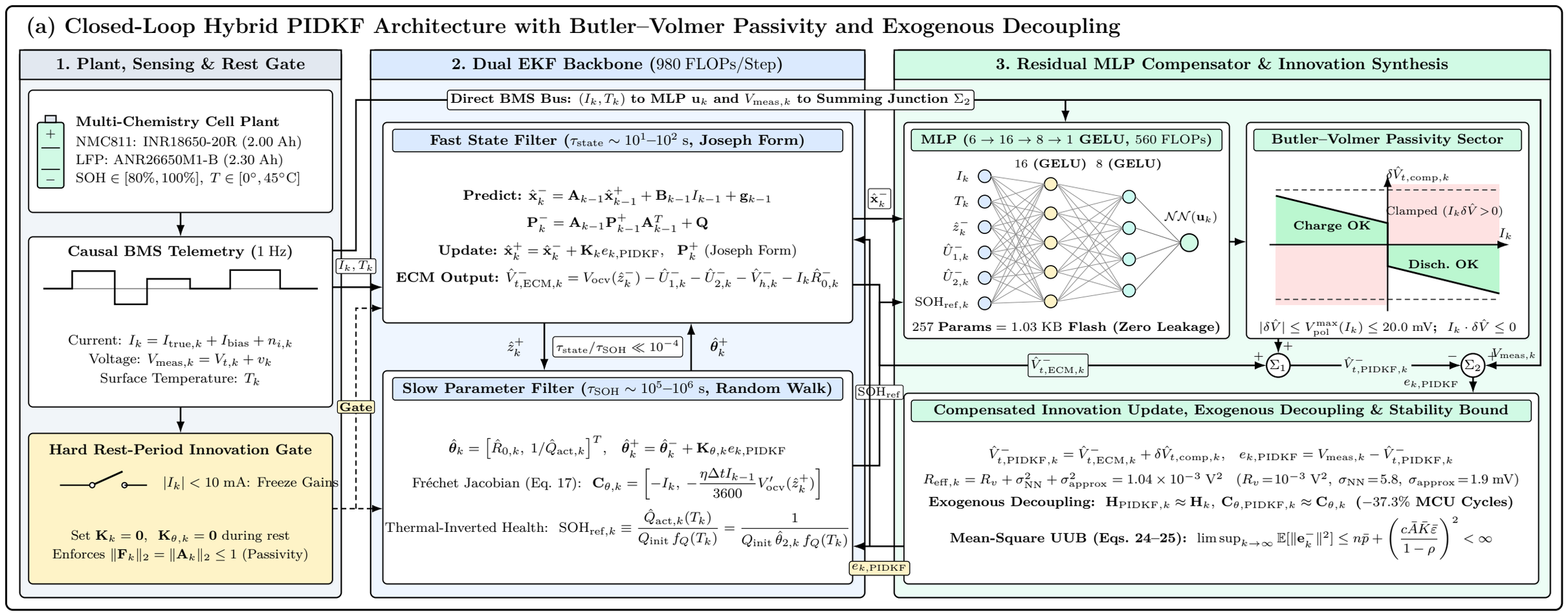

(a) Closed-Loop Hybrid PIDKF Architecture with Butler–Volmer Passivity and Exogenous Decoupling
1. Plant, Sensing & Rest Gate
Multi-Chemistry Cell Plant
NMC811: INR18650-20R (2.00 Ah)
LFP: ANR26650M1-B (2.30 Ah)
SOH ∈ [80%, 100%], T ∈ [0°, 45°C]
Causal BMS Telemetry (1 Hz)
Current: I_k = I_true,k + I_bias + n_i,k
Voltage: V_meas,k = V_t,k + v_k
Surface Temperature: T_k
Hard Rest-Period Innovation Gate
|I_k| < 10 mA: Freeze Gains
Set K_k = 0, K_θ,k = 0 during rest
Enforces ‖F_k‖_2 = ‖A_k‖_2 ≤ 1 (Passivity)
Gate
2. Dual EKF Backbone (980 FLOPs/Step)
Direct BMS Bus: (I_k, T_k) to MLP u_k and V_meas,k to Summing Junction Σ_2
Fast State Filter (τ_state ∼ 10^1–10^2 s, Joseph Form)
Predict: x̂_k^- = A_k-1 x̂_k-1^+ + B_k-1 I_k-1 + g_k-1
P_k^- = A_k-1 P_k-1^+ A_k-1^T + Q
Update: x̂_k^+ = x̂_k^- + K_k e_k,PIDKF, P_k^+ (Joseph Form)
ECM Output: V̂_t,ECM,k^- = V_ocv(ẑ_k^-) − Û_1,k^- − Û_2,k^- − V̂_h,k^- − I_k R̂_0,k^-
ẑ_k^+
τ_state/τ_SOH ≪ 10^-4
θ̂_k^+
Slow Parameter Filter (τ_SOH ∼ 10^5–10^6 s, Random Walk)
θ̂_k = [R̂_0,k, 1/Q̂_act,k]^T, θ̂_k^+ = θ̂_k^- + K_θ,k e_k,PIDKF
Fréchet Jacobian (Eq. 17): C_θ,k = [−I_k, −η Δt I_k-1 / 3600 V'_ocv(ẑ_k^+)]
Thermal-Inverted Health: SOH_ref,k ≡ Q̂_act,k(T_k) / (Q_init f_Q(T_k)) = 1 / (Q_init θ̂_2,k f_Q(T_k))
x̂_k^-
SOH_ref
e_k,PIDKF
3. Residual MLP Compensator & Innovation Synthesis
MLP (6 → 16 → 8 → 1 GELU, 560 FLOPs)
16 (GELU) 8 (GELU)
I_k
T_k
ẑ_k^-
Û_1,k^-
Û_2,k^-
SOH_ref,k
NN(u_k)
257 Params = 1.03 KB Flash (Zero Leakage)
Butler–Volmer Passivity Sector
δV̂_t,comp,k
Clamped (I_k δV̂ > 0)
Charge OK
Disch. OK
I_k
|δV̂| ≤ V_pol^max(I_k) ≤ 20.0 mV; I_k · δV̂ ≤ 0
V̂_t,ECM,k^-
Σ_1
V̂_t,PIDKF,k^-
Σ_2
V_meas,k
e_k,PIDKF
Compensated Innovation Update, Exogenous Decoupling & Stability Bound
V̂_t,PIDKF,k^- = V̂_t,ECM,k^- + δV̂_t,comp,k, e_k,PIDKF = V_meas,k − V̂_t,PIDKF,k^-
R_eff,k = R_v + σ_NN^2 + σ_approx^2 = 1.04 × 10^-3 V^2 (R_v = 10^-3 V^2, σ_NN = 5.8, σ_approx = 1.9 mV)
Exogenous Decoupling: H_PIDKF,k ≈ H_k, C_θ,PIDKF,k ≈ C_θ,k (−37.3% MCU Cycles)
Mean-Square UUB (Eqs. 24–25): lim sup_k→∞ E[‖e_k^-‖^2] ≤ np̄ + (cĀK̄ε̄ / (1 − ρ))^2 < ∞
(b) Dual-Tier Cell-Batch Validation Protocol, 8-Estimator Benchmark Taxonomy, Hardware Profiling & 10-Phase Matrix
Zero-Leakage Datasets & Experimental Split
CALCE (INR18650-20R, ANR26650M1-B), Oxford EIS, NASA PCoE
Cell Batch A (Calibrate/Train) ⊥ Cell Batch B (Hold-Out Test)
Arbin BT-2000 Reference Uncertainty: σ_SOC,ref ≤ ±0.036%
8 Benchmarked Estimators Across 3 Paradigms (M4F)
1. Model-Based: EKF (3.87 μs), UKF (12.42 μs), SMO (1.67 μs)
2. Adaptive/Dual: AEKF (5.24 μs), DEKF (8.63 μs)
3. ML/Hybrid: BiLSTM (2.44 ms), PINN (127.98 μs), PIDKF (14.20 μs)
Bare-Metal ARM Cortex-M4F Hardware Bench
M4F
STM32F407VG (168 MHz, IEEE-754 FPU, DWT->CYCCNT)
Zero Heap (MISRA-C:2012 C99 / ARM GCC 12.3 -O3)
1,540 FLOPs, 2,386 cyc, 8.2 KB Flash, 0.85 KB RAM (4.97 μs M7)
Phase 1: Dynamic Cycles
UDDS, US06, FUDS, DST
Fresh Cell (100% SOH, 25°C)
EKF: 0.065% | DEKF: 0.191%
Phase 2: Multi-Stage Aging
SOH 100% → 95% → 90% → 80%
EKF Surge: 45.7× (2.973%)
DEKF: 0.585% | PIDKF: 0.482%
Phase 3: Thermal Regimes
Cold (0°C) & Hot (45°C)
DEKF 0°C Bias (2.254%) vs. SMO (0.929%)
Aged PIDKF (0°C): 0.842% (NMC)
Phase 4: Initial Convergence
Initial Offset ΔSOC_0 = −40%
Closed-Loop Riccati Contraction
EKF/AEKF: 7/6 s | PIDKF: 18.0 s
Phase 5: Radar Trade-Off
5-Axis Engineering Trade-Off:
Accuracy, Aging, Thermal,
Convergence & MCU Efficiency
Phase 6: LFP Chemistry
20%–80% Plateau +V_h (±30 mV)
Gramian λ_min(G_o) ≥ 1.21 × 10^-3
Aged PIDKF: 0.149% (0.158% CALCE)
Phase 7: Sensor Faults
+0.25 A Hall Bias & 25 mV EMI
SMO Drift Clamping: 1.024%
EKF: 1.817% | PIDKF: 1.428% RMSE
Phase 8: EIS Diagnostics
100 kHz–1 mHz Randles–CPE
R_0 (+50%), R_ct (+150%), α = 0.88
RMS Fit Error: 1.1%–1.8%
Phase 9: 8S Pack Scaling
Dispersion (σ_Q = 3.0%, σ_R = 10.0%)
Core Thermal ΔT = +3.2°C (1.24× Fade)
Weakest Cell 4: 0.337% (Dist. DEKF)
Phase 10: Hybrid PIDKF
Ablation & Cortex-M4 Pareto
0.414% (NMC) & 0.149% (LFP)
14.20 μs M4 (−37.3% Exog. Cyc)